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OPERATOR QUICK REFERENCE GUIDE
ADCS TELEMETRY ANOMALY RESPONSE ? CORRECTIVE ACTIONS
ESA OPS-SAT Mission Operations
Version 1.0
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PART 1: CORRECTIVE ACTIONS MASTER TABLE
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This table provides the complete corrective action reference indexed by
anomaly type. Each action is graded by invasiveness (LOW = monitoring only,
MEDIUM = configuration change, HIGH = hardware command, CRITICAL = safe mode).

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ANOMALY ID | ANOMALY DESCRIPTION        | CORRECTIVE ACTIONS
-----------------------------------------------------------------------

MAG-AN-001 | Magnetometer X bias drift
           | CADC0872 mean shifted
           | Severity: MEDIUM
           | Channels: CADC0872
           |
           | Action 1 (LOW): Wait 15 min ? check if correlated with
           |   recent magnetorquer activity. Residual field decays.
           | Action 2 (MEDIUM): Compute bias from 5+ telemetry frames.
           |   Uplink CA-M01 (magnetometer bias correction telecommand).
           | Action 3 (MEDIUM): If temperature-correlated, adjust
           |   thermal model in ADCS calibration parameters.
           | Action 4 (HIGH): If bias >15 ?T and persistent: switch to
           |   star tracker primary (CA-M04).
           | Escalate: Engineering team if no improvement in 3 orbits.
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MAG-AN-002 | Magnetometer Y bias drift
           | CADC0873 mean shifted
           | Same actions as MAG-AN-001 applied to Y-axis.
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MAG-AN-003 | Magnetometer Z bias drift
           | CADC0874 mean shifted
           | Same actions as MAG-AN-001 applied to Z-axis.
-----------------------------------------------------------------------
MAG-AN-004 | All 3 magnetometers biased
           | CADC0872+0873+0874 all shifted
           | Severity: HIGH
           |
           | Action 1 (LOW): Verify orbital position vs IGRF model.
           |   Check geomagnetic storm status (Kp index).
           | Action 2 (MEDIUM): Uplink CA-M01 for all three axes.
           | Action 3 (MEDIUM): Cross-check EPS telemetry for power
           |   supply anomaly that could cause common-mode bias.
           | Action 4 (HIGH): If attitude determination quality metric
           |   is degraded, switch to star tracker primary (CA-M04).
           | Escalate: Immediately if Kp < 4 and bias unexplained.
-----------------------------------------------------------------------
MAG-AN-005 | Single magnetometer stuck value
           | One of CADC0872/0873/0874 constant, zero variance
           | Severity: HIGH
           |
           | Action 1 (MEDIUM): Send ADCS_MAG_RESET_CMD (CA-M02).
           |   Wait 2 minutes for sensor reinitialisation.
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           | Action 2 (HIGH): If soft reset fails: power cycle (CA-M03).
           |   Note: 30-60 second outage on that axis.
           | Action 3 (MEDIUM): While axis is failed, switch attitude
           |   estimation to 2-axis mag + star tracker if available.
           | Action 4 (MEDIUM): Update Kalman filter uncertainty to
           |   reflect degraded sensor state.
           | Escalate: If power cycle fails ? hardware fault confirmed.
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MAG-AN-006 | All 3 magnetometers stuck simultaneously
           | All channels constant, zero variance
           | Severity: CRITICAL
           |
           | Action 1 (HIGH): Power cycle ADCS controller module.
           | Action 2 (CRITICAL): If power cycle fails: initiate
           |   safe mode (CA-S01) to prevent attitude loss.
           | Action 3: Contact ADCS engineering team immediately.
           | Note: This is extremely rare. Most likely OBC software
           |   crash with stale telemetry being reported.
-----------------------------------------------------------------------
MAG-AN-007 | Magnetometer high noise (EMI correlated)
           | Elevated variance during known EMI activity
           | Severity: LOW (expected)
           |
           | Action 1 (LOW): Confirm magnetorquers are active.
           |   This is EXPECTED during torquing operations.
           | Action 2 (LOW): Schedule magnetometer sampling windows
           |   during torquer-off intervals (standard practice).
           | Action 3 (LOW): Log event. No intervention needed if
           |   noise clears within 2 minutes of torquer deactivation.
-----------------------------------------------------------------------
MAG-AN-008 | Magnetometer high noise (no known EMI source)
           | Elevated variance, no explanation from activity logs
           | Severity: HIGH
           |
           | Action 1 (MEDIUM): Check all experiment payload status.
           |   Identify any drawing pulsed power.
           | Action 2 (MEDIUM): If experiment is causing EMI:
           |   command experiment pause via OBC.
           | Action 3 (MEDIUM): Enable increased averaging (5-sample
           |   running average) via ADCS configuration command.
           | Action 4 (HIGH): If noise makes attitude determination
           |   unreliable: switch to star tracker primary (CA-M04).
           | Escalate: If persists > 3 orbits with no identified cause.
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MAG-AN-009 | Out-of-range magnetometer reading
           | Any channel exceeds ±65 ?T
           | Severity: MEDIUM
           |
           | Action 1 (LOW): Verify orbital position ? near poles,
           |   up to 60 ?T is normal.
           | Action 2 (LOW): Check magnetorquer drive current ? very
           |   high current levels can saturate sensor.
           | Action 3 (MEDIUM): If software scaling error suspected
           |   after recent ADCS firmware update: re-uplink calibration
           |   table with correct scaling coefficients.
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PD-AN-001  | Single photodiode dark (non-eclipse)
           | One of CADC0875-0880 reads zero when it should be lit
           | Severity: MEDIUM
           |
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           | Action 1 (LOW): Cross-reference with current attitude
           |   and sun vector. Is this face expected to be dark?
           | Action 2 (LOW): Check eclipse prediction. Is the satellite
           |   near eclipse or just exiting? Transition takes time.
           | Action 3 (MEDIUM): Command small attitude offset (5 deg)
           |   to test if channel responds ? helps distinguish sensor
           |   failure from pointing anomaly.
           | Action 4 (MEDIUM): If confirmed failure: CA-P01 (exclude
           |   channel from sun vector calculation).
           | Monitor: Sun vector accuracy with 5 remaining channels.
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PD-AN-002  | All 6 photodiodes dark (not predicted eclipse)
           | All CADC0875-0880 near zero, out of eclipse window
           | Severity: CRITICAL
           |
           | Action 1 (MEDIUM): Immediately verify eclipse prediction
           |   using orbital mechanics tool ? confirm this is not a
           |   delayed or extended eclipse event.
           | Action 2 (CRITICAL): If satellite should be in sunlight:
           |   attitude may be wrong (satellite pointed wrong way) or
           |   major power/sensor anomaly has occurred.
           | Action 3 (CRITICAL): Check power subsystem for total power
           |   failure ? if EPS battery voltage is very low, satellite
           |   may be in power-critical safe mode.
           | Action 4 (CRITICAL): Initiate safe mode (CA-S01) ? this
           |   will command a sun search using B-dot + photodiode logic.
           | Escalate: Immediate emergency contact with operations team.
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PD-AN-003  | Photodiode saturation (stuck at maximum)
           | One channel reads ADC maximum value
           | Severity: LOW to MEDIUM
           |
           | Action 1 (LOW): Check if Sun is directly face-on to this
           |   sensor ? high irradiance can naturally saturate sensor.
           | Action 2 (MEDIUM): If not explained by geometry: exclude
           |   channel via CA-P01 to prevent corrupting sun vector.
           | Action 3 (LOW): Check ADC reference voltage in ADCS HK.
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PD-AN-004  | Photodiode noisy / inconsistent sun vector
           | Multiple channels showing inconsistent pattern
           | Severity: MEDIUM to HIGH
           |
           | Action 1 (HIGH PRIORITY): Check ADCS rate sensor.
           |   Rapid rates indicate TUMBLING. If tumbling: CA-S01.
           | Action 2 (MEDIUM): If rates are nominal: increase photodiode
           |   averaging to 10 samples per sun vector computation.
           | Action 3 (MEDIUM): If inconsistency persists: switch to
           |   star tracker primary for attitude reference.
           | Action 4 (LOW): Review experiment activity for EMI coupling
           |   to analog photodiode circuits.
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PD-AN-005  | Photodiode gradual degradation
           | Progressive decrease in peak readings over weeks
           | Severity: LOW (but escalating)
           |
           | Action 1 (LOW): Confirm with attitude cross-check: perform
           |   calibration pass at known sun angle to measure sensitivity.
           | Action 2 (MEDIUM): Update photodiode gain table via CA-P02.
           | Action 3 (LOW): Monitor rate of degradation. Log monthly.
           | Action 4 (MEDIUM): If >30% sensitivity loss: increase sun
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           |   vector averaging; document for mission lifetime model.
           | Action 5 (HIGH at >50% loss): Switch attitude determination
           |   to star tracker primary; use photodiodes for safe mode only.
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MULTI-001  | Both magnetometers AND photodiodes anomalous
           | Mixed anomalies across all ADCS channels
           | Severity: CRITICAL
           |
           | Action 1: IMMEDIATELY check ADCS angular rate sensors.
           |   If rates > 2 deg/s: satellite is tumbling ? CA-S01 NOW.
           | Action 2: Check EPS battery voltage and current. Total power
           |   failure produces anomalies on all analog channels.
           | Action 3: If power is low: enter minimum-power safe mode.
           | Action 4: If power is nominal but attitude is lost:
           |   initiate safe mode ? B-dot detumble + sun search.
           | Action 5: Mobilise full operations and engineering team.
           | Note: This pattern is the most serious ADCS anomaly scenario.
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PART 2: DECISION TREES FOR OPERATOR USE
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DECISION TREE A: MAGNETOMETER ANOMALY RESPONSE

    Is the satellite in a geomagnetic storm (Kp > 5)?
    YES ? Monitor only. Increase IGRF uncertainty. No corrective action.
    NO  ?

    Were magnetorquers active in the past 15 minutes?
    YES ? This is likely residual EMI. Wait 15 min and reassess.
    NO  ?

    Is the anomaly a stuck/frozen value?
    YES ? Is it on 1 axis? YES ? CA-M02 (soft reset)
                           NO  ? CA-M03 (power cycle) ? CA-S01 if fails
    NO  ?

    Is it a bias (mean shifted, normal variance)?
    YES ? CA-M01 (bias correction). Monitor. Escalate if no improvement.
    NO  ?

    Is it high noise?
    YES ? Check experiment activity. Pause EMI source. Enable averaging.
    NO  ?

    Escalate to engineering team.


DECISION TREE B: PHOTODIODE ANOMALY RESPONSE

    Is the satellite in predicted eclipse?
    YES ? All PDs dark is NORMAL. Monitor for eclipse exit.
    NO  ?

    Is only one photodiode channel dark?
    YES ? Is the satellite pointing that face away from Sun?
          YES ? NORMAL. No action.
          NO  ? CA-P01 (exclude channel). Investigate cause.
    NO  ?
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Are all six photodiodes dark outside eclipse?
YES ? CRITICAL: Check power system. Initiate CA-S01.
NO ?

Are multiple channels noisy/inconsistent?
YES ? Check ADCS rate sensors first.
      High rates ? CA-S01 (detumble).
      Normal rates ? Increase averaging. Switch to star tracker.
NO ?

Is one channel saturated?
YES ? Check sun geometry. If not explained: CA-P01.
NO ?

Are all channels showing gradual long-term decrease?
YES ? CA-P02 (calibration update). Log degradation trend.

PART 3: COMMUNICATION TEMPLATES FOR ANOMALY REPORTING

When reporting an anomaly to the engineering team, use the following format:

ANOMALY REPORT TEMPLATE:

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Date/Time UTC: [YYYY-MM-DD HH:MM:SS]
Orbit Number: [###]
Orbital Phase: [Sunlit / Eclipse / Eclipse entry / Eclipse exit]
Channels Affected: [list CADC#### channels]
Anomaly Type: [Point / Contextual / Collective / Drift]
Anomaly Pattern: [describe observed deviation]
Duration: [time since first detection]
Cross-Subsystem Checks:
  - Magnetorquer status at anomaly time: [active / inactive]
  - EPS status: [nominal / anomalous]
  - ADCS rate sensors: [rates in deg/s]
  - Experiment payload status: [list active payloads]
Space Weather: [Kp index, any storms]
Root Cause Hypothesis: [most likely cause with confidence level]
Corrective Actions Taken: [list actions and results]
Current Status: [resolved / ongoing / monitoring]
Recommended Next Steps: [actions for next contact window]
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PART 4: GLOSSARY OF TECHNICAL TERMS

ADCS: Attitude Determination and Control System
ADC: Analog-to-Digital Converter
B-DOT: Rate-of-change of magnetic field; used by B-dot de-tumble controller
CADC: Telemetry channel prefix for OPS-SAT ADCS channels
EMI: Electromagnetic Interference
EPS: Electrical Power System
FDIR: Fault Detection, Isolation and Recovery
IGRF: International Geomagnetic Reference Field ? standard Earth field model
I2C: Inter-Integrated Circuit ? communication bus used to read sensor data
Kp Index: Geomagnetic activity index (0=quiet, 9=severe storm)
LEO: Low Earth Orbit
OBC: On-Board Computer

OBCP: On-Board Control Procedure
OOL: Out-Of-Limits alarm
PUS: Packet Utilisation Standard ? ESA satellite communication protocol
RCA: Root Cause Analysis
RW: Reaction Wheel
SAA: South Atlantic Anomaly ? radiation belt region
SEU: Single-Event Upset ? bit flip caused by cosmic ray
TID: Total Ionising Dose ? cumulative radiation damage
?T: Microtesla ? unit of magnetic field strength

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END OF OPERATOR QUICK REFERENCE GUIDE
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